

Design, Modeling, and Experimental Validation of a Batch Distillation System for Micro-Distillery Ethanol Production

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ABSTRACT

This study presents the design, modeling, and experimental validation of a batch distillation system for traditional micro-distilleries operating at high altitudes. The theoretical framework incorporated altitude-adjusted boiling points and vapor-liquid equilibrium modeling using Antoine and Margules equations. A prototype with an LPG-fired reboiler and a four-stage column with copper bubble-cap trays was fabricated and tested in three trials with varying feed compositions and volumes. Reflux ratios ranged from 0.625 to 1.164, and energy input from 1.02 to 1.60 kW. The model showed strong agreement with experimental results, predicting distillate ethanol concentrations with relative errors between 0.08% and 3.2% v/v, confirming the reliability of the pressure-corrected thermodynamic framework for optimizing small-scale ethanol production.

Keywords: Batch distillation, vapor-liquid equilibrium, experimental validation, distillation apparatus, mathematical modeling.

I. INTRODUCTION

Distillation is one of the oldest and most widely applied separation processes, used in industries ranging from petroleum refining to alcoholic beverage production. For traditional micro-distilleries, especially in developing regions such as Nepal, designing cost-effective and efficient units is important for supporting local economies and sustainable technologies. Many existing systems are built empirically, without rigorous mathematical validation or consideration of site-specific conditions such as high altitude, where reduced atmospheric pressure changes boiling behavior.

Batch distillation is particularly suitable for small-scale alcohol production, offering operational flexibility. However, accurate prediction of vapor-liquid equilibrium (VLE), estimation of energy requirements, and optimization of column performance remain key challenges. While several studies have investigated these issues theoretically, limited work has been carried out to link mathematical modeling with practical testing at the micro-distillery scale.

This study aims to bridge that gap by integrating mathematical modeling, computational simulation, and experimental validation. Boiling points of ethanol and water were adjusted for altitude using barometric and Clausius–Clapeyron relations. VLE behavior was modeled with the Antoine and Margules equations. Column performance was analyzed using the McCabe–Thiele method, supported by energy and mass balance calculations. A prototype apparatus was designed and fabricated at Pulchowk Campus, and experimental trials were performed to evaluate heating time, reflux ratio, and distillate composition. The comparison between predicted and measured results showed close agreement, confirming the applicability of the proposed design for small-scale ethanol production under high-altitude conditions.

This study makes three main contributions. First, it develops a mathematical framework to predict boiling points, vapor-liquid equilibrium, reflux ratio, and stage-by-stage composition, completing the design of the distillation column. Second, it presents a prototype built from locally available materials, suitable for small traditional micro-distilleries. Third, the predictions from the model are confirmed through experimental testing.

II. RELATED WORK

Distillation has been extensively studied in both industrial and academic contexts, with emphasis on large-scale chemical processing. Classical works such as McCabe and Thiele’s method provide a foundation for graphical and analytical design of distillation columns [1]. Subsequent advances have introduced rigorous mathematical models based on material and energy balances, vapor-liquid equilibrium (VLE) thermodynamics, and numerical simulation methods [2], [3]. These approaches have enabled optimization of column efficiency, energy consumption, and separation performance in large-scale operations.

For ethanol–water systems, several models have been applied to predict VLE behavior, including Raoult’s law, activity coefficient models such as Wilson, NRTL, and UNIQUAC, and empirical correlations [4], [5]. Studies have shown that the Margules equation provides a reasonable balance of accuracy

and simplicity for binary ethanol–water mixtures, particularly under moderate pressure conditions. cubic spline fitting techniques have also been employed to approximate equilibrium curves for computational efficiency [6].

In the context of small-scale alcohol production, most distillation units are constructed empirically without extensive modeling or optimization. Research has reported that micro-distilleries often suffer from inefficiencies such as excessive fuel consumption, uncontrolled reflux ratios, and inconsistent product purity [7], [8]. Few works, however, have combined rigorous thermodynamic modeling with the practical design and fabrication of apparatus intended for traditional or rural applications.

Recent studies have attempted to bridge this gap by introducing hybrid approaches that integrate simulation with experimental validation. For example, laboratory-scale distillation studies using simplified equilibrium models have been validated against experimental runs for binary systems [9]. Others have explored the influence of altitude and ambient pressure on boiling characteristics and distillation efficiency, particularly in mountainous regions [10]. These works emphasize the need for localized design parameters to ensure process efficiency.

Building upon these efforts, this study contributes by applying high-altitude corrected thermodynamic modeling, polynomial-based VLE approximation, and McCabe–Thiele analysis to a practically fabricated apparatus for micro-distillery applications. Unlike prior works that focus solely on theoretical or experimental aspects, this work integrates both, thereby providing a validated framework for efficient small-scale ethanol production in high-altitude environments.

III. MATHEMATICAL MODELING

The performance of the batch distillation apparatus was analyzed using a thermodynamic framework consisting of pressure and boiling point adjustment, vapor-liquid equilibrium (VLE) modeling, energy balance for power estimation and reflux ratio determination. Detailed derivations of the following equations are provided in Appendix A.

A. Atmospheric Pressure and Boiling Point Adjustment

Because the experiments were carried out in Kathmandu (altitude ~ 1300 m), the local atmospheric pressure differs from standard sea-level conditions. The pressure at the testing site was calculated using the barometric relation (Eq. (A.1)), which gave a value of 652.42 mmHg (or 86982.35 Pa). Using this pressure in the Clausius-Clapeyron relation (Eq. (A.2)), the corresponding boiling temperatures were obtained for the two components. The corrected boiling points are approximately 74.28°C for ethanol and 95.71°C for water.

B. Vapor-Liquid Equilibrium (VLE)

The vapor-liquid equilibrium for the ethanol–water system was evaluated using Raoult’s law with non-ideal activity coefficients:

$$y_i P = x_i \gamma_i P_i^{\text{sat}}(T), \quad (1)$$

where x_i and y_i denote the liquid- and vapor-phase mole fractions, γ_i is the activity coefficient, and $P_i^{\text{sat}}(T)$ is the saturation pressure of component i . Saturation pressures were calculated using the Antoine relation, and liquid-phase non-ideality was represented with the two-parameter Margules model.

C. Energy Balance and Power Input

Direct measurement of the heating provided by the LPG stove is challenging due to heat losses to the surroundings. To account for this, we estimated the effective heat supplied by recording the time required for the feed to reach its boiling point. The effective heating power was calculated from the experimental heating data using:

$$P_{\text{supplied}} = \frac{(T_{\text{bpt}} - T_0) [\rho_e C_{e,0} c_{p,e} + \rho_w (1 - C_{e,0}) c_{p,w}] V_0}{\Delta t_d} \quad (2)$$

where T_{bpt} is the boiling point of water, T_0 is the initial temperature of the feed, $C_{e,0}$ is the initial ethanol concentration, V_0 is the feed volume, ρ denotes density, c_p is the specific heat, and Δt_d is the heating duration.

The heat removed by the reflux condenser is:

$$Q_{\text{cond}} = \dot{m}_{\text{cw}} c_p (T_{\text{out}} - T_{\text{in}}), \quad (3)$$

where $\dot{m}_{\text{cw}}$ is the mass flow rate of cooling water (kg/s), c_p is the specific heat of water (J/kg·K), and T_{in} and T_{out} are the inlet and outlet temperatures of the reflux cooling water (°C).

D. Reflux Ratio

The reflux ratio R was obtained from condenser and reboiler energy balances, leading to:

$$R = \frac{Q_{\text{cond}}}{P_{\text{supplied}} - Q_{\text{cond}}}, \quad (4)$$

where Q_{cond} is reflux condenser heat duty and P_{supplied} is the supplied heating power.

E. McCabe–Thiele Column Analysis

Column performance was analyzed using the McCabe–Thiele graphical method. The rectifying operating line is given by:

$$y = \frac{R}{R+1}x + \frac{x_D}{R+1}, \quad (5)$$

where x_D is distillate composition.

The stage-wise construction established a functional relationship between the bottoms composition (x_W) and the distillate composition (x_D) for a given reflux ratio, expressed as $x_W(x_D)$, which was subsequently inverted to $x_D(x_W)$. This relationship was implemented using cubic spline interpolation (Fig.11) to enable efficient simulation.

The instantaneous distillate composition was then determined as a function of the decreasing bottoms composition over time. The rate of change in bottom composition and

quantity depends on the heat supplied and the reflux ratio, and the process is tracked step by step for the selected time period.

F. Step-wise Calculation

Before performing the step-wise mass and composition calculations, the total simulation duration was determined based on the experimental heating time. The simulation only tracks the process after the feed reaches its boiling point (t_{bpt}). The effective simulation time and time-stepping were defined as:

$$\text{Simulation_time} = t_{\text{total}} - t_{\text{bpt}}, \quad \Delta t_{\text{step}} = 1 \text{ second}$$

$$N_{\text{steps}} = \frac{60 \times \text{Simulation_time}}{\Delta t_{\text{step}}} \quad (6)$$

where, t_{total} is the total experimental duration in minutes, Simulation_time is the post-boiling simulation period (minutes), Δt_{step} is the time step size, and N_{steps} is the total number of simulation steps.

At each step, the bottoms mole fraction, distillate withdrawal, composition, and cumulative volumes are updated as described below. The liquid composition in the boiler is first updated from the remaining moles of ethanol and water. The bottoms mole fraction of ethanol is

$$x_W^{(k)} = \frac{n_{e,\text{remain}}^{(k-1)}}{n_{e,\text{remain}}^{(k-1)} + n_{w,\text{remain}}^{(k-1)}} \quad (7)$$

The corresponding distillate composition is then obtained from the cubic-spline map derived from the McCabe-Thiele analysis,

$$x_D^{(k)} = f(x_W^{(k)}) \quad (8)$$

Based on the energy input during the time interval Δt , the moles of distillate withdrawn are

$$n_{\text{dist}} = \frac{P \Delta t}{(1 + R) [x_D M_{w,e} L_{v,e} + (1 - x_D) M_{w,w} L_{v,w}]} \quad (9)$$

where R is the reflux ratio.

The moles of ethanol and water collected in that step are

$$\Delta n_e = x_D n_{\text{dist}}, \quad \Delta n_w = (1 - x_D) n_{\text{dist}}. \quad (10)$$

The cumulative moles collected up to step k are

$$n_e^{(k)} = \sum_{i=1}^k \Delta n_e^{(i)}, \quad n_w^{(k)} = \sum_{i=1}^k \Delta n_w^{(i)}$$

The corresponding masses are

$$M_e^{(k)} = n_e^{(k)} M_{w,e}$$

$$M_w^{(k)} = n_w^{(k)} M_{w,w}$$

$$M_t^{(k)} = M_e^{(k)} + M_w^{(k)}$$

The ethanol weight percentage of the collected distillate is

$$w_e^{(k)} = \frac{M_e^{(k)}}{M_t^{(k)}} \times 100. \quad (11)$$

To determine the density of the ethanol-water mixture, a cubic-spline interpolation $S(w_e)$ was constructed from tabulated density data at 20°C [11]. The mixture density at each time step is

$$\rho_{\text{mix}}^{(k)} = S(w_e^{(k)}) \quad (12)$$

The mixed liquid volume accumulated up to step k , accounting for contraction upon mixing, is

$$V_{\text{mix}}^{(k)} = \frac{M_t^{(k)}}{\rho_{\text{mix}}^{(k)}} \quad (13)$$

Instantaneous and Cumulative Volume: At each step k , the pure-component volumes collected during that interval are obtained from the incremental masses of ethanol and water,

$$\Delta V_e^{(k)} = \frac{M_e^{(k)} - M_e^{(k-1)}}{\rho_e}; \quad \Delta V_w^{(k)} = \frac{M_w^{(k)} - M_w^{(k-1)}}{\rho_w} \quad (14)$$

where ρ_e and ρ_w are the densities of pure ethanol and water, respectively.

The cumulative collected volumes up to step k are then

$$V_e^{(k)} = \sum_{i=1}^k \Delta V_e^{(i)}, \quad V_w^{(k)} = \sum_{i=1}^k \Delta V_w^{(i)}$$

Instantaneous and Cumulative Concentration: The cumulative ethanol concentration (v/v%) in the distillate at step k is

$$C_{\text{cumu}}^{(k)} = \frac{V_e^{(k)}}{V_e^{(k)} + V_w^{(k)}} \times 100$$

The instantaneous ethanol concentration (v/v%) for interval k is obtained from the incremental volume collected during that step,

$$C_{\text{inst}}^{(k)} = \frac{\Delta V_e^{(k)}}{\Delta V_e^{(k)} + \Delta V_w^{(k)}} \times 100$$

These instantaneous and cumulative quantities were compared with the experimental measurements for model validation. Because it is difficult to precisely control the reflux ratio, heat input, and condenser heat duty, experimental data were first collected and then used as input for the mathematical model.

IV. DESIGN AND FABRICATION OF THE APPARATUS

Based on the parameters derived from the mathematical modeling, a prototype apparatus was designed and fabricated to validate the theoretical framework. The prototype batch distillation unit was designed in SolidWorks and fabricated locally.

A. Schematic Design

A schematic representation of the batch distillation system is shown in Fig. 1. The setup consists of five main components: the boiler (or reboiler), the distillation column with bubble cap trays, the reflux condenser, the main product condenser, and the distillate collection system.

In operation, the feed is heated in the boiler, generating ethanol–water vapor. The vapor rises through the column, where staged separation occurs via bubble cap trays that enhance vapor–liquid contact. A portion of the vapor is condensed in the reflux condenser and returned to the column to improve separation efficiency, while the remaining vapor passes to the main condenser for complete liquefaction. The condensed distillate is finally collected and analyzed for concentration and volume.

The schematic highlights the overall flow of heat and mass in the apparatus and illustrates the interaction between the reboiler, separation stages, and condensers, providing a clear overview of the system before detailed component design.

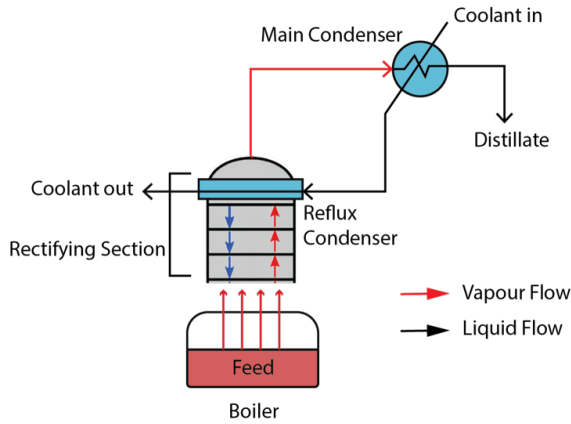


Fig. 1: Schematic diagram of the batch distillation apparatus showing major components and process flow.

B. Overall Design

The apparatus is made up of several key components: the boiler (or reboiler/feed container), the distillation column, the condenser system, a set of bubble cap trays, and the collection system.

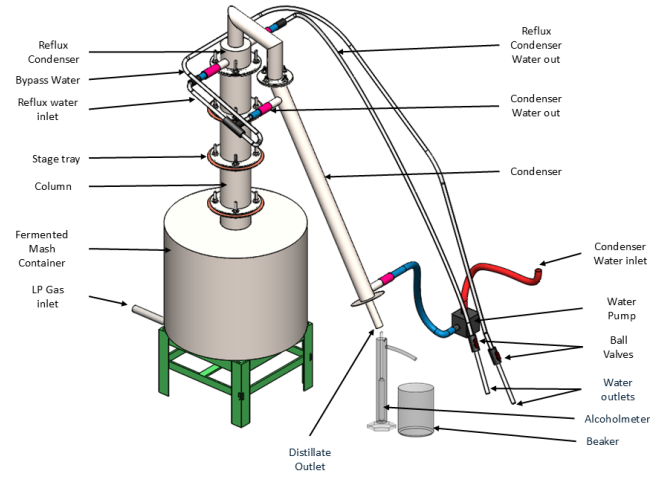


Fig. 2: Overview of the designed batch distillation apparatus.

The boiler is made of stainless steel and holds about 29 liters. It is designed to be heated by an LPG burner, but can also work with induction, infrared, or biofuel. The distillation column has space for four theoretical stages, including three bubble cap trays and the boiler stage, based on the McCabe–Thiele analysis. The condenser is a water-cooled shell-and-tube type, with a controlled water supply for inlet and outlet. The distillate is collected in calibrated glassware, allowing the volume and composition to be measured during experiments.

C. Boiler and Heating System

The boiler was fabricated from food-grade stainless steel to prevent corrosion and contamination. An LPG burner was selected as the heat source due to local availability and cost-effectiveness. Heating power was indirectly estimated from the heating rate of the feed, as described in Eq. (2). The 3D design of boiler is shown in Fig. 3.

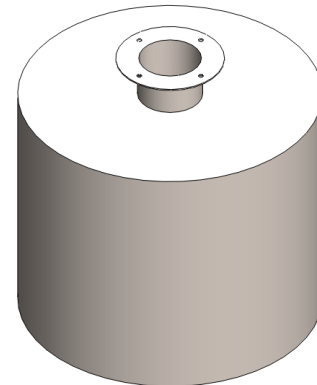


Fig. 3: Boiler.

D. Distillation Column

The distillation column was designed for modular operation, providing up to four theoretical equilibrium stages. It was

fabricated from stainless steel to ensure durability, resistance to corrosion, and suitability for food-grade applications. The overall height and diameter of the column were determined based on McCabe–Thiele analysis, optimized to achieve efficient separation while maintaining ease of fabrication and assembly. Figure 4 presents the CAD model of a column.

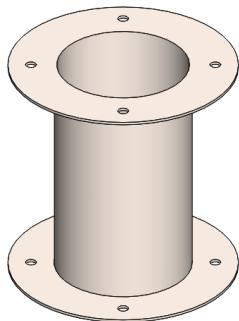


Fig. 4: CAD model of the stainless steel distillation column designed for four equilibrium stages.

E. Bubble Cap Trays

To facilitate vapor–liquid contact, three bubble cap trays were incorporated inside the column, in addition to the reboiler stage. Each tray consisted of two risers and a downcomer, designed to maintain liquid holdup while allowing vapor passage through the caps. The trays were fabricated from copper, chosen for its high thermal conductivity and traditional use in alcohol distillation. Trays were installed between the column sections using silicone gaskets for sealing and secured with stainless steel nut–bolt fasteners. The CAD model of the tray assembly is shown in Fig. 5.

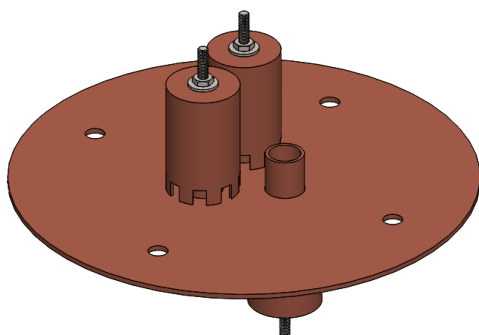


Fig. 5: CAD model of the copper bubble cap tray assembly with risers and downcomer.

F. Condenser and Cooling Water System

Two types of condensers were integrated into the apparatus: a reflux condenser and a main product condenser. The reflux condenser employed a cross-flow configuration, where cooling water flowed horizontally while ethanol vapors rose vertically through the column. In contrast, the main condenser utilized a counter-flow arrangement, with ethanol vapor passing through

the inner tube and cooling water circulating in the outer jacket in the opposite direction. Both units were fabricated using stainless steel tubing to ensure durability and efficient heat transfer.

Cooling water flow rate and temperature rise at the inlet and outlet were recorded to calculate the condenser duty using Eq. (3). These measurements were further applied in determining the reflux ratio. The condenser design also allowed for adjustable water flow rates, enabling sensitivity studies on reflux conditions. CAD models of the reflux and main condensers are shown in Figs. 6 and 7.

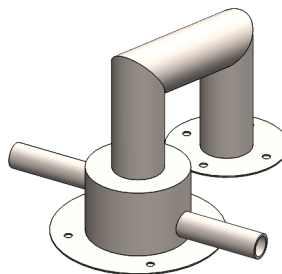


Fig. 6: CAD model of the reflux condenser with cross-flow configuration.

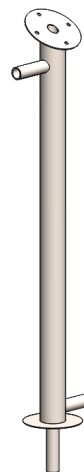


Fig. 7: CAD model of the main condenser with counter-flow configuration.

G. Instrumentation and Measurement

The apparatus was equipped with a set of sensors and measurement devices to monitor its operation. Thermocouples were installed at the boiler, column, and condenser outlets to track temperature variations throughout the process. Distillate volume was measured using graduated glass cylinders, while ethanol concentration was determined using an alcohol hydrometer (ABV method).

Cooling water flow rates were obtained using a volumetric calibration method, where the time required to fill a known

volume was recorded. Each experimental run was monitored for heating duration, onset of boiling, and the period of distillate collection, ensuring accurate characterization of the system's thermal and separation performance.

H. Fabrication Summary

The apparatus was fabricated using locally available materials and workshop facilities, with a total construction period of approximately two weeks. Stainless steel was employed for the column and boiler to ensure strength, corrosion resistance, and suitability for food-grade operation, while copper was selected for the bubble cap trays to enhance heat transfer and maintain traditional distillation practices. Standard fastening elements and silicone gaskets were used to provide modular assembly and leak-tight operation.

The completed prototype demonstrated structural stability and reliability during repeated heating-cooling cycles, confirming the robustness of the design for small-scale alcohol production. A photograph of the assembled batch distillation unit is shown in Fig. 8.

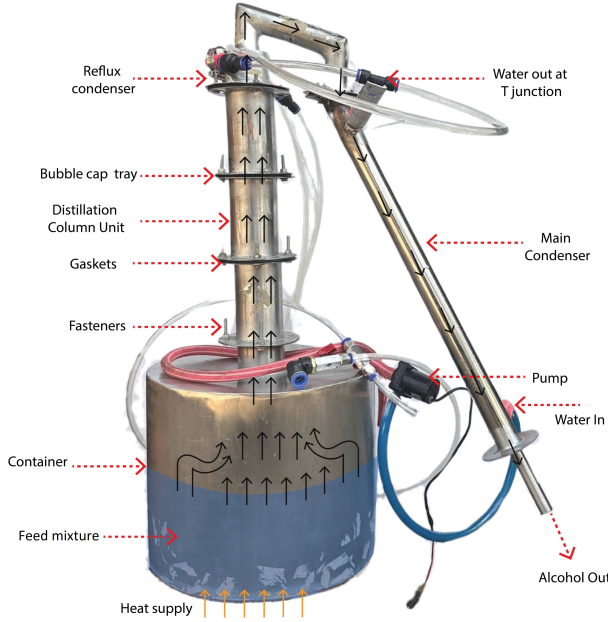


Fig. 8: Fabricated prototype of the batch distillation apparatus, showing the boiler, column, condenser system, and collection unit.

V. EXPERIMENTAL SETUP AND PROCEDURE

To validate the theoretical model and assess the performance of the fabricated apparatus, a series of controlled experimental trials were conducted at Pulchowk Campus laboratory. Each run involved charging the reboiler with a known quantity of feed, initiating heating with LPG fuel, and recording key process parameters over time.

A. Feed Preparation and Experimental Runs

Three experimental runs were conducted using different feed compositions, prepared sequentially.

Run 1: The feed consisted of fermented mash containing suspended solids and heterogeneous components. It was filtered before charging, although minor inconsistencies in composition were unavoidable.

Run 2: The feed was prepared by diluting the distillate recovered from Run 1 with additional ethanol obtained from separate test runs.

Run 3: The feed was formulated by combining ethanol recovered from Run 2 with extra ethanol from other test runs, followed by dilution to the desired starting concentration.

B. Operating Conditions and Input Parameters

The experiments were conducted under ambient conditions in Kathmandu (altitude 1300 m) using a distillation column with four theoretical stages. To ensure full reproducibility of the experimental and numerical results, all operating, environmental, and initial conditions used in the three experimental runs are summarized in Table I. Where multiple measurements were recorded during a run, time-averaged values are reported. These parameters were used directly as inputs to the thermodynamic and dynamic models without further tuning.

TABLE I: Operating and environmental conditions for experimental runs

Parameter	Run 1	Run 2	Run 3
Feed and Initial Conditions			
Initial feed volume (L)	4.0	3.0	3.5
Initial ethanol concentration (v/v%)	20	27	30
Initial feed temperature (°C)	18.3	15.1	17.0
Operating Conditions			
Total experimental duration (min)	40	38	53
Time to reach boiling point (min)	12.00	14.08	15.67
Ambient temperature (°C)	19.1	17.0	18.0
Condenser and Cooling Water			
Cooling-water inlet temperature (°C)	25.0	20.7	24.73
Cooling-water outlet temperature (°C)	29.1	31.95	37.1
Cooling-water flow rate (mL/s)	36.00	11.69	9.71

C. Instrumentation and Measurements

During each trial, the following measurements were recorded:

- **Heating time:** Time required to reach the boiling point (onset of distillate collection), along with multiple intervals for measuring distillate volume and composition.
- **Cooling water:** Inlet and outlet temperatures, as well as volumetric flow rate.
- **Distillate properties:** Volume collected at specified intervals and alcohol content, determined using an alcohol hydrometer. Real-time measurements were averaged over short time intervals.

D. Procedure

At the beginning of each trial, the feed temperature was recorded. Heating was then initiated, and the time required to reach the boiling point was noted. Once distillate collection commenced, samples were periodically withdrawn and analyzed for alcohol concentration. Simultaneously, the inlet and outlet temperatures of the condenser water, as well as the water flow rate, were recorded. The data collected from these measurements were subsequently used as input for the mathematical model.

VI. RESULTS AND DISCUSSION

This section presents the results of the mathematical modeling, Python simulations, and experimental trials, followed by a comparison and interpretation of the findings. The discussion focuses on the accuracy of the theoretical predictions, the stability of the apparatus during operation, and the implications for small-scale ethanol production.

A. Energy Balance and Reflux Ratio values

In these three experimental runs, the supplied heating power was calculated using Eq. (2), and the condenser duty was determined from the measured cooling-water flow rate and temperature rise using Eq. (3). The reflux ratio for each run was then obtained from Eq. (4). A summary of the results from all three tests is given in Table II.

TABLE II: Summary of energy balance results for all experimental runs

Parameter	Run 1	Run 2	Run 3
Supplied heating power (W)	1602.49	1020.92	1025.87
Condenser heat duty (W)	616.32	549.07	501.53
Reflux ratio (R)	0.625	1.164	0.956

B. Vapor-Liquid Equilibrium (VLE) Curve

The vapor-liquid equilibrium behavior of the ethanol-water system was obtained from the Antoine and Margules relations under the local atmospheric conditions of Kathmandu. Fig. 9 shows the equilibrium curve predicted by the Antoine-Margules model.

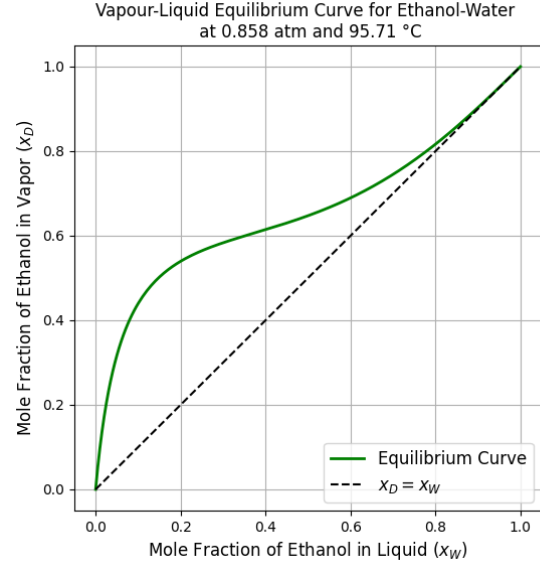


Fig. 9: Vapor-liquid equilibrium (VLE) curve for the ethanol-water system at testing site condition

C. Column Performance (McCabe-Thiele Analysis)

The McCabe-Thiele method was applied to predict the stage-wise performance of the distillation column. Using four theoretical stages and the experimentally determined reflux ratio, the relationship between the bottoms (x_W) and distillate (x_D) compositions was obtained and fitted with a cubic spline for computational smoothness. Fig. 11 illustrates the spline-fitted x_W - x_D relationship, and Fig. 10 presents the corresponding McCabe-Thiele diagram. One of the point is chosen for visualization.

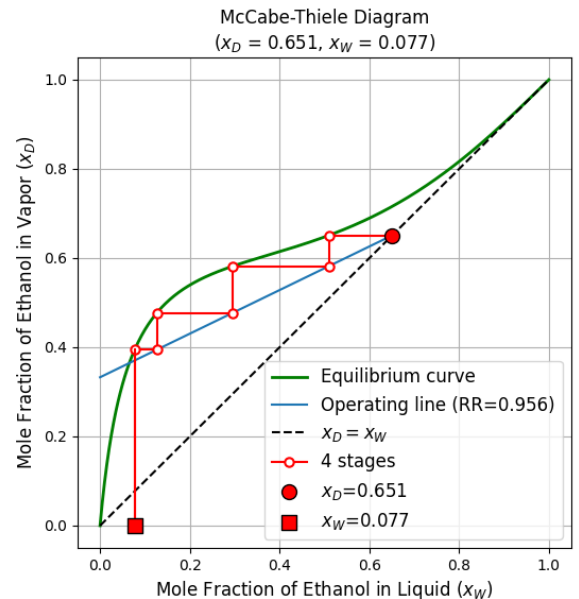


Fig. 10: McCabe-Thiele diagram for the ethanol-water system (4 stages, $R = 0.956$) for Run 3.

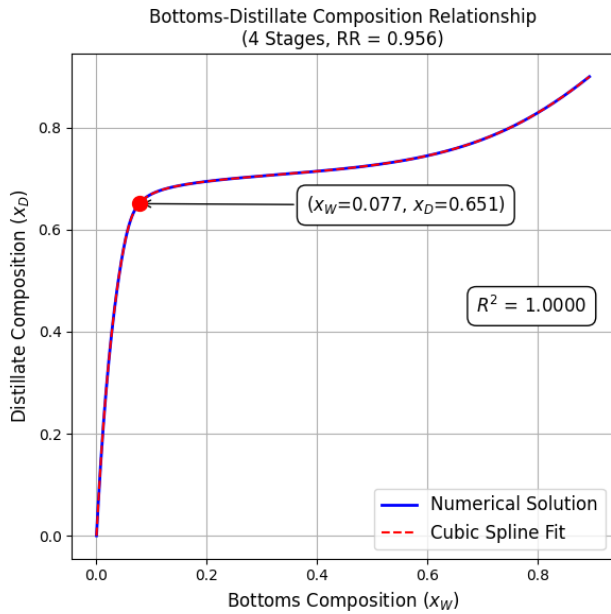


Fig. 11: Bottoms composition vs. distillate composition for 4 stages at reflux ratio $R = 0.956$ for Run 3.

A set of discrete distillate compositions was generated with finer resolution near low-purity values to accurately capture the steep changes in the McCabe–Thiele stepping procedure. This non-uniform spacing ensured smooth and accurate cubic spline interpolation of the $x_D(x_W)$ relationship.

D. Distillate Profiles

The mathematical model was used to simulate the temporal evolution of distillate collection and composition during batch operation. For illustration, the predicted trends for Run 3 are presented in three representative plots: cumulative distillate volume versus time, cumulative distillate composition tracking, instantaneous ethanol concentration versus time, and instantaneous distillate volume collected per minute. Time zero corresponds to the start of heating; the plot begins when the feed reaches its boiling point and the first distillate is collected.

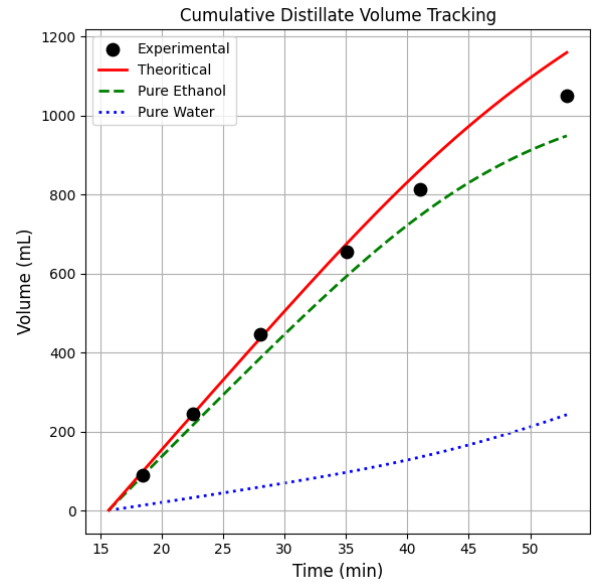


Fig. 12: Cumulative distillate volume collected as a function of time for Run 3.

The ethanol concentration (v/v%) was then determined from the ratio of pure ethanol volume to the total volume of the pure components. However, the cumulative distillate volume was calculated using the ethanol weight fraction and the corresponding mixture density from Eq. 13.

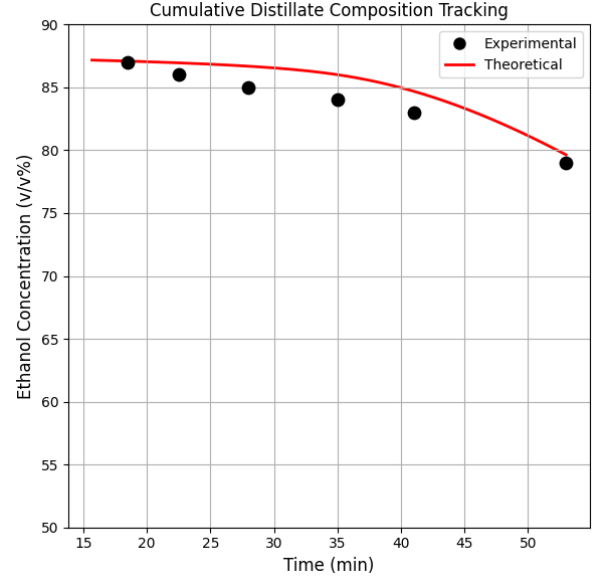


Fig. 13: Cumulative distillate Composition collected as a function of time for Run 3.

Distillate was collected in a graduated cylinder until an approximately 90 mL volume was accumulated, and the elapsed collection time was recorded. The collected distillate volume was divided by the corresponding collection time to obtain an average distillate rate, which was reported on a per-minute

basis. Ethanol concentration was measured using a hydrometer for each collected sample, and the corresponding data point was assigned to the midpoint of the associated time interval for temporal representation.

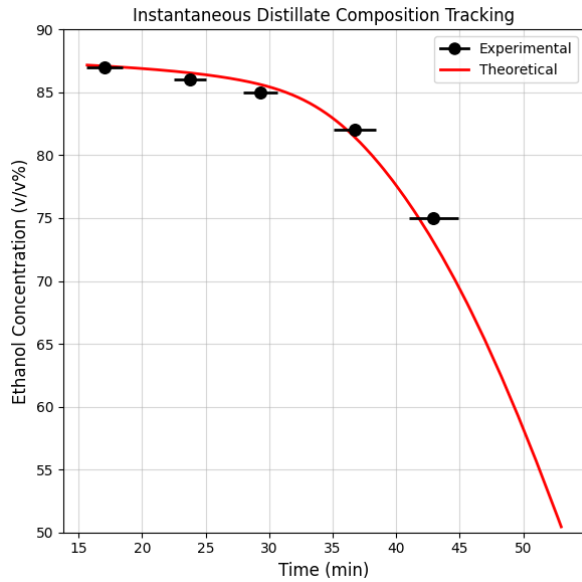


Fig. 14: Instantaneous distillate ethanol concentration as a function of time for Run 3.

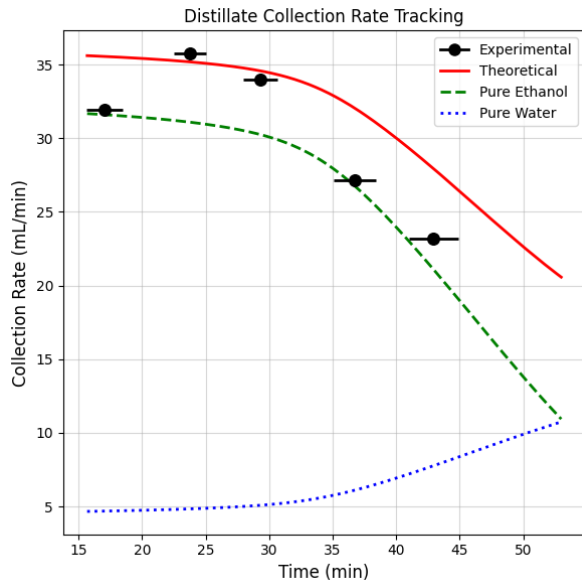


Fig. 15: Instantaneous distillate volume collected per minute for Run 3.

The cumulative volume profile (Figure 12) reflects the progressive recovery of vapor condensate over time, while the ethanol concentration curve (Figure 14) shows the gradual decline in purity typical of batch distillation. The instantaneous distillate rate (Figure 15) highlights the changing vapor gen-

eration rate as the ethanol content in the reboiler decreases. The experimental data and mathematical model show excellent agreement, as evident in the overlapping trends. The model successfully captures the characteristic decline in distillate ethanol concentration over time, corresponding to the gradual depletion of volatile components in the boiler.

E. Comparison of Experimental and Predicted Results

Table III summarizes the measured and predicted values of the final distillate ethanol concentration and total distillate volume for all three runs.

TABLE III: Comparison of experimental and model-predicted results

Runs	Run 1	Run 2	Run 3
Ethanol Concentration (v/v%)			
Theoretical	59.92	84.07	79.64
Experimental	58	84	79
Abs. Error	1.92	0.07	0.64
% Error	3.20%	0.08%	0.81%
Distillate Volume (mL)			
Theoretical	1228.76	722.14	1159.37
Experimental	1182	593	1049
Abs. Error	46.76	129.14	110.37
% Error	3.81%	17.88%	9.52%

F. Discussion of Findings

The combined theoretical and experimental analysis demonstrates that the developed modeling framework is applicable to batch ethanol–water distillation under high-altitude operating conditions. Across all experimental runs, the model reproduced the principal trends observed during operation and provided quantitatively consistent predictions of column behavior. The main findings of the study can be summarized as follows:

- The predicted temporal evolution of distillate ethanol concentration showed close agreement with experimental measurements, with small absolute and relative deviations throughout the batch duration.
- The experimental apparatus exhibited stable and repeatable operation under varying feed compositions and operating conditions, indicating its suitability for controlled small-scale batch distillation studies.
- The energy-based estimation of heating power and reflux ratio proved sufficient to capture the dominant process dynamics without the need for direct vapor or liquid flow measurements.

Although the model demonstrated strong predictive performance, discrepancies between predicted and experimental results can be attributed to a combination of physical system limitations and measurement or modeling assumptions.

a) *Systematic Sources of Error:*

- **Unaccounted heat losses:** Heat losses due to radiation, natural convection, and imperfect insulation reduce the effective thermal energy available for vapor generation. These losses are not explicitly included in the power estimation, leading to deviations in predicted vapor production rates.
- **Variability in cooling-water conditions:** Temporal fluctuations in cooling-water inlet temperature and flow rate influence the condenser heat duty and, consequently, the instantaneous reflux ratio. Time-averaged values of the measured cooling-water temperatures and flow rates were used in the energy balance calculations.
- **Measurement uncertainties:** Instrumental limitations, including the resolution of the measuring cylinder and alcoholmeter, along with manual timing and observation delays, introduce uncertainty in the recorded distillate volume and composition.
- **Operational losses:** Vapor leakage, condensate holdup, and small losses during distillate collection result in deviations from ideal mass and energy balances.

b) *Run-Specific Observations:*

- **Run 1:** The presence of suspended solids in the feed led to irregular boiling behavior and non-uniform vapor release. This affected the estimation of effective heating power and introduced variability in the measured distillate collection rate.
- **Run 2:** A localized leakage at the bubble-cap joint reduced effective vapor–liquid contact within the column, resulting in a lower-than-predicted distillate recovery.
- **Run 3:** Residual liquid retained on the trays at startup diluted the initial vapor composition, causing minor deviations during the early stages of the batch operation.

Despite these practical limitations, the close overall agreement between experimental observations and model predictions confirms that the integrated thermodynamic and energy-based framework provides a reliable basis for the analysis and design of batch distillation systems for micro-distillery applications under high-altitude conditions.

VII. CONCLUSION

This work presented the development, modeling, and experimental validation of a small-scale batch distillation system for ethanol–water separation, motivated by applications in traditional and micro-distillery settings. A consistent theoretical framework was established, incorporating altitude-adjusted boiling point estimation, vapor–liquid equilibrium modeling with activity coefficients, energy-based determination of heating input, and indirect evaluation of the reflux ratio. Column behavior was analyzed using the McCabe–Thiele method, and the resulting stagewise relationships were embedded into a dynamic simulation through spline-based interpolation.

A laboratory-scale apparatus was designed and fabricated using locally available materials, and three experimental runs were conducted under different initial feed compositions and

operating conditions. The effective heating power and condenser heat duty were inferred from measured temperature histories and cooling-water data, enabling estimation of the reflux ratio without direct flow measurement. These experimentally derived inputs were used directly in the mathematical model, without parameter tuning.

The simulated trends in distillate composition, and cumulative distillate volume showed close agreement with experimental observations. In particular, the model captured the characteristic decline in ethanol concentration during batch operation, reflecting progressive depletion of the volatile component in the reboiler. The results demonstrate that the proposed energy-based modeling approach provides a physically consistent and practically accurate description of batch distillation performance.

Overall, the integration of experimental measurements with equilibrium-based modeling offers a reliable framework for analyzing small-scale batch distillation systems. The methodology developed in this study provides a sound basis for future extensions, including improved reflux control strategies, energy-efficiency analysis, and adaptation to other binary mixtures relevant to decentralized distillation applications.

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APPENDIX

APPENDIX A: MATHEMATICAL DERIVATIONS

This appendix presents the detailed derivations of the equations used in the modeling and analysis of the batch distillation apparatus. The main body of the paper presents only the final governing relations; here, the intermediate steps are provided for completeness and reproducibility.

A. Atmospheric Pressure at Altitude

Atmospheric pressure decreases with altitude h according to the barometric relation:

$$P(h) = P_0 \exp\left(-\frac{gM_{\text{air}}h}{R_gT}\right), \quad (\text{A.1})$$

where P_0 is the sea-level pressure, g is gravitational acceleration, M_{air} is the molar mass of air, R_g is the universal gas constant, and T is ambient temperature in Kelvin.

B. Boiling Point Correction (Clausius–Clapeyron)

The shift in boiling point with pressure was obtained using the Clausius–Clapeyron relation:

$$\ln\left(\frac{P_2}{P_1}\right) = -\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right), \quad (\text{A.2})$$

where P_1, P_2 are pressures at temperatures T_1, T_2 respectively, and ΔH_{vap} is the enthalpy of vaporization. This was used to compute the altitude-corrected boiling temperatures of ethanol and water.

C. Vapor–Liquid Equilibrium (VLE)

For the ethanol–water system, the equilibrium vapor composition was computed using Raoult’s law with activity coefficients:

$$y_i P = x_i \gamma_i P_i^{\text{sat}}(T), \quad (\text{A.3})$$

where x_i and y_i denote the liquid- and vapor-phase mole fractions, γ_i is the activity coefficient, and $P_i^{\text{sat}}(T)$ is the temperature-dependent saturation pressure of component i .

Antoine Equation:

$$\log_{10} [P_i^{\text{sat}}(T)] = A_i - \frac{B_i}{T + C_i}, \quad (\text{A.4})$$

where T is in °C and P_i^{sat} is in mmHg. The Antoine constants for the ethanol–water system are:

Ethanol: $A = 8.20417, B = 1642.89, C = 230.3$

Water: $A = 8.07131, B = 1730.63, C = 233.426$

Margules Activity Coefficient Model: Liquid-phase non-ideality was represented using the two-parameter Margules equations:

$$\ln \gamma_e = x_w^2 [A_{ew} + 2x_e(A_{we} - A_{ew})], \quad (\text{A.5})$$

$$\ln \gamma_w = x_e^2 [A_{we} + 2x_w(A_{ew} - A_{we})], \quad (\text{A.6})$$

where x_e and x_w are the liquid mole fractions of ethanol and water, respectively. The parameters A_{ew} and A_{we} are empirical interaction coefficients obtained from ethanol–water VLE measurements. Common literature values used in this work are:

$$A_{ew} = 1.6022, \quad A_{we} = 0.7947.$$

Vapor Mole Fraction Expression: Combining Raoult’s law, the Antoine equation, and the Margules model yields the vapor-phase mole fraction of ethanol:

$$y_e = \frac{\gamma_e x_e P_e^{\text{sat}}}{\gamma_e x_e P_e^{\text{sat}} + \gamma_w x_w P_w^{\text{sat}}}. \quad (\text{A.7})$$

This expression was evaluated pointwise to create VLE Curve.

D. Reflux Ratio Derivation

The reflux ratio was determined from the energy balance between the condenser and the reboiler.

The heat removed by the condenser is:

$$Q_{\text{cond}} = \dot{m}_{\text{cw}} c_p (T_{\text{out}} - T_{\text{in}}), \quad (\text{A.8})$$

where $\dot{m}_{\text{cw}}$ is the mass flow rate of cooling water (kg/s), c_p is the specific heat of water (J/kg·K), and T_{in} and T_{out} are the inlet and outlet temperatures of the reflux cooling water (°C).

The reboiler supplies energy to vaporize the liquid mixture:

$$P_{\text{supplied}} = n_D (1 + R) H, \quad (\text{A.9})$$

where n_D is the molar flow rate of distillate (mol/s), R is the reflux ratio, and H is the average enthalpy required to vaporize one mole of the liquid mixture, accounting for the composition and latent heats of the components (J/mol) given as:

$$H = x_D M_e L_e + (1 - x_D) M_w L_w \quad (\text{A.10})$$

Combining these expressions, the reflux ratio is calculated as:

$$R = \frac{Q_{\text{cond}}}{P_{\text{supplied}} - Q_{\text{cond}}}. \quad (\text{A.11})$$

APPENDIX B: CODE REPOSITORY

The Python scripts and related materials for simulations, equilibrium curve generation, and McCabe-Thiele analysis are available online at:

GitHub Repository: Batch Distillation Project